

Content

Preface

9758 H2 Mathematics Syllabus (checklist for topics)

List of Formulae MF27

- 1 Equations and Inequalities
 - 2 Differentiation Techniques
 - 3 Integration Techniques
 - 4 Sequences and Series, AP and GP
 - 5 Complex Numbers
 - 6 Vectors 1 (Vector Algebra, Ratio Theorem, Scalar and Vector Product)
 - 7a Vectors 2 (Lines and Planes)
 - 7b Vectors 3 (Lines and Planes)
 - 8 Applications of Differentiation
 - 9 Graphing Techniques and Transformations
 - 10 Functions
 - 11 Applications of Integration
 - 12 Differential Equations
-
- 13 Permutations and Combinations, Probability
 - 14 Discrete Random Variable, Binomial Distribution, Normal Distribution
 - 15 Sampling and Estimations, Hypothesis Testing, Correlation & Regression
-
- 16 Review the Use of Graphing Calculators

Preface: TIPS for H2 Mathematics Revision

You should first get your notes and materials organised.

Make sure you have the following:

	Purpose	Remarks
<u>Revision Booklet</u> <u>Topical Summary</u>	First check on understanding of topic	If there are parts of the summary alien to you, consider revisiting lecture notes or videos. You may refer to 9758 H2 Mathematics syllabus document for a checklist of key concepts required for each topic.
<u>Practice Questions</u>	If you want exposure to various question types for topics you feel you wish to get stronger in.	Recommended to try these questions, or at least employ the strategy suggested under Tutorials and Assignments if you have insufficient time.
<u>Lecture Notes</u>	If you are completely clueless on a topic, you need to start here.	This is the bare minimum you should ideally prepare for.
<u>Tutorials and Assignments</u>	If you are not that confident about a topic and need a quick refresher, start here.	
<u>Additional Practice</u>	If you want to get exposure to various question types for each topic (these were given as each topic was discussed, so some of you may already have completed these).	
<u>Practice Papers</u>	When all your topical work is done, and you are all ready to take on the 3 hour challenge.	Strongly recommended to try at least 1 paper from 8am to 11am before the actual exam sitting.

You would want to consider 2 different kinds of revision sessions:

- 1) **Topical (1.5-2 hours)**
- 2) **Practice Papers (3 hours)**

Topical (We recommend 1.5 hours for each topic)

Start with a topic that you have revised before or are confident in first. Try to do 2 questions from the Additional Practice or the Revision Booklet in the first 20 minutes. It is a good practice to continually retrieve knowledge, no matter how confident you are. This helps you to

1. Reinforce Neural Pathways

Retrieving information (even "easy" content) strengthens long-term memory.

2. Identify Hidden Gaps

Confidence $\neq$ mastery.

Compare your worked solutions with those presented. Have you missed out any key steps, explanations? Consider compiling/annotating on the revision booklet what you should pay attention to.

After that, you will focus the remaining hour for the topic that you wish to strengthen on. For that topic, one possible approach is the following:

- a) For 30 minutes, use the summary inside the revision booklet to make sure that you are familiar with what is required. If there are concepts that you are not sure of, refer to the Lecture Notes for more detailed explanation and further examples.
- b) For the remaining 30 minutes, you should try to do 2 questions from the Additional Practice or Revision Booklet itself.

Key Principles to Emphasize

1. 50% Rule: Spend *at least half the session solving problems* — math is a skill, not a spectator sport (so this is a gentle reminder not to just read solutions, as this won't help you retain the takeaways).
2. Spaced Repetition: Revisit the same topic in *3 days* to lock in learning.

Practice Papers (3 hours)

- You need to carve out a continuous 3 hour sitting for it after you have finished your revision for all the chapters. This should preferably be 8am to 11am or 1pm to 4pm, same as the actual paper.
- Simulate actual conditions by having your GC, MF27 [print from SEAB website, do not use a digital version] and stationery in front of you. Print the question paper as well for the best simulation.
- Focus on time management. You need to move on from a question/part of question if you are using more than 2 times the number of marks allocated in terms of minutes spent. Whilst it is onerous at first to keep track of time (what time you started on a part of question), as you get more comfortable, you should have a good feel of when you are overspending time on a question.
- Focus on stamina. Stay seated (that is your primary position during exams) and focused, only going to the washroom if there is a need to. All these add up in a 3-hour paper. Try not to plug in music and stay focused no matter what kind of environment you are in.
- At the end of the 3 hours, check your work against the solutions and see where you made mistakes and learn from them. Do not bother giving marks or thinking how many marks you will get. If there were parts you were stuck in, look back at the syllabus requirements for the chapter as well as the revision booklet and see if you have missed out an important concept, and add on to your notes there.
- Of course, seek your tutors for consults, especially if you are not sure about your solution.

GC Matters

- For questions that do not require exact answers, you should know how to use your GC to efficiently obtain them. These include
 - Using PolySmlt 2 (system of linear equations, polynomial equations)
 - Finding intersection of curves (to solve equations)
 - Calculating min and max points of curves
 - Using the table of values to solve problems involving least integer/greatest integer satisfying certain conditions
 - Using parametric mode and sequence mode appropriately if the question requires it
 - Various statistics functions (Stat Plot, Edit lists, 1 Var-Stat, LinReg, etc)
 - Shortcuts like alpha + any of the first row buttons
- Don't skip through the GC screenshots in your lecture notes and go through the motions of the various sequences of button pushes! This will help you to gain familiarity with the GC.
- Finally, ensure the GC is fully charged before examinations.

YOU CAN DO IT! BELIEVE IN YOURSELF, CONFIDENCE IS KEY! 🍀

**Singapore–Cambridge General Certificate of Education
Advanced Level Higher 2 (2025)
Mathematics
(Syllabus 9758)**

(First year of examination in 2025)

extracted from https://www.seab.gov.sg/files/A%20Level%20Syllabus%20Sch%20Cddts/2025/9758_y25_sy.pdf

SCHEME OF EXAMINATION PAPERS

For the examination in H2 Mathematics, there will be two 3-hour papers, each carrying 50% of the total mark, and each marked out of 100, as follows:

PAPER 1 (3 hours)

A paper consisting of 10 to 12 questions of different lengths and marks based on the Pure Mathematics section of the syllabus.

There will be one question on application of Mathematics in real-world contexts, including those from sciences and engineering. This question will carry at least 12 marks and may require concepts and skills from more than one topic.

Candidates will be expected to answer **all** questions.

PAPER 2 (3 hours)

A paper consisting of two sections, Sections A and B.

Section A (Pure Mathematics – 40 marks) will consist of 4 to 5 questions of different lengths and marks based on the Pure Mathematics section of the syllabus.

Section B (Probability and Statistics – 60 marks) will consist of 6 to 8 questions of different lengths and marks based on the Probability and Statistics section of the syllabus.

There will be one question in Section B on application of Mathematics in real-world contexts, including those from sciences and engineering. This question will carry at least 12 marks and may require concepts and skills from more than one topic.

Candidates will be expected to answer **all** questions.

CONTENT OUTLINE

Knowledge of the content of the O-Level Mathematics syllabus is assumed. The assumed knowledge for O-Level Additional Mathematics is appended after this section.

	Topic/Sub-topics	Content
SECTION A: PURE MATHEMATICS		
1	Functions and graphs	
1.1	Functions	Include: ☐ concepts of function, domain and range ☐ inverse functions and composite functions ☐ conditions for the existence of inverse functions and composite functions ☐ domain restriction to obtain an inverse function ☐ relationship between graphs of a one-to-one function and its inverse Exclude the use of the relation $(fg)^{-1} = g^{-1}f^{-1}$, and restriction of domain to obtain a composite function.
1.2	Graphs and transformations	Include: ☐ use of a graphing calculator or a graphing software to graph a given function ☐ important characteristics of graphs such as symmetry, intersections with the axes, turning points and asymptotes of the following: $y^2 = ax ; x^2 = by$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 ; \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ $y = \frac{ax + b}{cx + d}$ $y = \frac{ax^2 + bx + c}{dx + e}$ ☐ equations of asymptotes, axes of symmetry, and restrictions on the possible values of x and/or y ☐ effect of transformations on the graph of $y = f(x)$ as represented by $y = af(x)$, $y = f(x) + a$, $y = f(x + a)$ and $y = f(ax)$ and combinations of these transformations ☐ relating the graphs of $y = f(x)$, $y = f(x)$, and $y = \frac{1}{f(x)}$ to the graph of $y = f(x)$ ☐ simple parametric equations and their graphs

	Topic/Sub-topics	Content
1.3	Equations and inequalities	Include: ☐ formulating an equation, a system of linear equations, or inequalities from a problem situation ☐ solving an equation exactly or approximately using a graphing calculator <i>or a graphing software</i> ☐ solving a system of linear equations using a graphing calculator <i>or a graphing software</i> ☐ solving inequalities of the form $\frac{f(x)}{g(x)} > 0$ where $f(x)$ and $g(x)$ are linear expressions or quadratic expressions ☐ concept of x, and use of relations $x-a < b \Leftrightarrow a-b < x < a+b$ and $x-a > b \Leftrightarrow x < a-b$ or $x > a+b$ ☐ solving inequalities by graphical methods
2	Sequences and series	
2.1	Sequences and series	Include: ☐ concepts of sequence and series for finite and infinite cases ☐ sequence as function $y = f(n)$ where n is a positive integer ☐ relationship between u_n (the nth term) and S_n (the sum to n terms) ☐ sequence given by a formula for the nth term ☐ sequence generated by the relation $u_{n+1} = f(u_n)$, including the use of a graphing calculator <i>or a computer</i> to generate the sequence ☐ sum and difference of two series ☐ convergence of a series and the sum to infinity ☐ formula for the nth term and the sum of a finite arithmetic series ☐ formula for the nth term and the sum of a finite geometric series ☐ condition for convergence of an infinite geometric series ☐ formula for the sum to infinity of a convergent geometric series
3	Vectors	
3.1	Basic properties of vectors in two and three dimensions	Include: ☐ addition and subtraction of vectors, multiplication of a vector by a scalar, and their geometrical interpretations ☐ position vectors, displacement vectors and direction vectors ☐ magnitude of a vector ☐ unit vectors ☐ distance between two points ☐ collinearity

	Topic/Sub-topics	Content
		☐ use of the ratio theorem in geometrical applications
3.2	Scalar and vector products in vectors	Include: ☐ concepts of scalar product and vector product of vectors and their properties ☐ angle between two vectors ☐ geometrical meanings of $\mathbf{a} \cdot \hat{\mathbf{n}}$ and $\mathbf{a} \times \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is a unit vector Exclude triple products $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ and $\mathbf{a} \times \mathbf{b} \times \mathbf{c}$.
3.3	Three-dimensional vector geometry	Include: ☐ vector and cartesian equations of lines and planes ☐ foot of the perpendicular and distance from a point to a line or to a plane ☐ angle between two lines, between a line and a plane, or between two planes ☐ relationships between (i) two lines (coplanar or skew) (ii) a line and a plane (iii) two planes Exclude: • shortest distance between two skew lines • common perpendicular to two skew lines
4	Introduction to Complex numbers	
4.1	Complex numbers expressed in cartesian form and Argand diagrams	Include: ☐ extension of the number system from real numbers to complex numbers ☐ complex roots of quadratic equations ☐ modulus, argument and conjugate of a complex number ☐ four operations of complex numbers ☐ equality of complex numbers ☐ conjugate roots of a polynomial equation with real coefficients ☐ representation of complex numbers in the Argand diagram ☐ geometrical effects of conjugation, negation, addition, subtraction, and multiplication by i Exclude complex numbers expressed in polar (or modulus-argument) form and exponential form.

	Topic/Sub-topics	Content
5	Calculus	
5.1	Differentiation	Include: ☐ graphical interpretation of (i) $f'(x) > 0$, $f'(x) = 0$ and $f'(x) < 0$ (ii) $f''(x) > 0$ and $f''(x) < 0$ ☐ relating the graph of $y = f'(x)$ to the graph of $y = f(x)$ ☐ differentiation of simple functions defined implicitly or parametrically ☐ determining the nature of the stationary points (local maximum and minimum points and points of inflexion) analytically, in simple cases, using the first derivative test or the second derivative test ☐ locating maximum and minimum points using a graphing calculator or a graphing software ☐ finding the approximate value of a derivative at a given point using a graphing calculator or a graphing software ☐ problems involving tangents and normals to curves, including cases where the curve is defined implicitly or parametrically ☐ local maxima and minima problems ☐ connected rates of change problems Exclude non-stationary points of inflexion and finding second derivative of functions defined parametrically.
5.2	Maclaurin series	Include: ☐ standard series expansion of $(1+x)^n$ for any rational n, e^x, $\sin x$, $\cos x$ and $\ln(1+x)$ ☐ derivation of the first few terms of the Maclaurin series by – repeated differentiation, e.g. $\sec x$ – repeated implicit differentiation, e.g. $y^3 + y^2 + y = x^2 - 2x$ – using standard series, e.g. $e^x \cos 2x$, $\ln\left(\frac{1+x}{1-x}\right)$ ☐ range of values of x for which a standard series converges ☐ concept of Maclaurin's series as an approximation of a function ☐ small angle approximations: $\sin x \approx x$, ☐ $\cos x \approx 1 - \frac{1}{2}x^2$, $\tan x \approx x$ Exclude problems involving derivation of the general term of a series.

	Topic/Sub-topics	Content
5.3	Integration techniques	Include: ☐ integration of $f'(x)[f(x)]^n$ (including $n = -1$), $f'(x)e^{f(x)}$ $\sin^2 x$, $\cos^2 x$, $\tan^2 x$ $\frac{1}{a^2 + x^2}$, $\frac{1}{\sqrt{a^2 - x^2}}$, $\frac{1}{a^2 - x^2}$ and $\frac{1}{x^2 - a^2}$ ☐ integration by a given substitution ☐ integration by parts Exclude reduction formulae.
5.4	Definite integrals	Include: ☐ concept of definite integral as a limit of sum ☐ definite integral as the area under a curve ☐ evaluation of definite integrals ☐ area of a region bounded by a curve and lines parallel to the coordinate axes, between a curve and a line, or between two curves ☐ area below the x-axis ☐ volume of revolution about the x- or y-axis ☐ finding the approximate value of a definite integral using a graphing calculator or a graphing software Exclude area and volume of revolution about the x-axis or y-axis where curve is defined parametrically.
5.5	Differential equations	Include: ☐ solving for the general solutions and particular solutions of differential equations of the form $\frac{dy}{dx} = f(x)g(y)$, including reducing a given differential equation to this form by means of a given substitution ☐ formulating a differential equation from a problem situation ☐ interpreting a differential equation and its solution in terms of a problem situation

	Topic/Sub-topics	Content
SECTION B: PROBABILITY AND STATISTICS		
6	Probability and Statistics	
6.1	Probability	Include: ☐ addition and multiplication principles for counting ☐ concepts of permutation (nP_r) and combination (nC_r) ☐ arrangements of objects in a line or in a circle, including cases involving repetition and restriction ☐ addition and multiplication of probabilities ☐ mutually exclusive events and independent events ☐ use of tables of outcomes, Venn diagrams, tree diagrams, and permutations and combinations techniques to calculate probabilities ☐ calculation of conditional probabilities in simple cases ☐ use of: $P(A') = 1 - P(A)$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A B) = \frac{P(A \cap B)}{P(B)}$
6.2	Discrete random variables	Include: ☐ concept of discrete random variables, probability distributions, expectations and variances ☐ concept of binomial distribution $B(n, p)$ as an example of a discrete probability distribution and use of $B(n, p)$ as a probability model, including conditions under which the binomial distribution is a suitable model ☐ use of mean and variance of binomial distribution (without proof) Exclude finding cumulative distribution function of a discrete random variable.

	Topic/Sub-topics	Content
6.3	Normal distribution	Include: ☐ concept of continuous random variables* ☐ concept of a normal distribution as an example of a continuous probability model and its mean and variance; use of $N(\mu, \sigma^2)$ as a probability model ☐ standard normal distribution ☐ finding the value of $P(X < x_1)$ or a related probability, given the values of x_1, μ, σ ☐ symmetry of the normal curve and its properties ☐ finding a relationship between x_1, μ, σ given the value of $P(X < x_1)$ or a related probability ☐ solving problems involving the use of $E(aX + b)$ and $\text{Var}(aX + b)$ ☐ solving problems involving the use of $E(aX + bY)$ and $\text{Var}(aX + bY)$, where X and Y are independent Exclude normal approximation to binomial distribution.
6.4	Sampling	Include: ☐ concepts of population and simple random sample ☐ concept of the sample mean $\bar{X}$ as a random variable with $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ ☐ distribution of sample mean from a normal population ☐ use of the Central Limit Theorem to treat sample mean as having normal distribution when the sample size is sufficiently large (e.g. $n \geq 30$) ☐ use of unbiased estimates of the population mean and variance from a sample, including cases where the data are given in summarised form $\sum x$ and $\sum x^2$, or $\sum(x - a)$ and $\sum(x - a)^2$

* For teaching and learning only.

	Topic/Sub-topics	Content
6.5	Hypothesis testing	Include: ☐ concepts of null hypothesis (H_0) and alternative hypotheses (H_1), test statistic, critical region, critical value, level of significance, and p-value ☐ formulation of hypotheses and testing for a population mean based on – a sample from a normal population of known variance – a large sample from any population ☐ 1-tail and 2-tail tests ☐ interpretation of the results of a hypothesis test in the context of the problem Exclude the use of the term 'Type I error', concept of Type II error and testing the difference between two population means.
6.6	Correlation and linear regression	Include: ☐ use of scatter diagram to judge if there is a plausible linear relationship between the two variables ☐ correlation coefficient as a measure of the fit of a linear model to the scatter diagram ☐ interpreting the product moment correlation coefficient (in particular, values close to -1, 0 and 1) ☐ concepts of linear regression and method of least squares to find the equation of the regression line ☐ concepts of interpolation and extrapolation ☐ use of the appropriate regression line to make prediction or estimate a value in practical situations, including explaining how well the situation is modelled by the linear regression model ☐ use of a square, reciprocal or logarithmic transformation to achieve linearity Exclude: • problems involving derivation of formulae • relationship $r^2 = b_1 b_2$, where b_1 and b_2 are regression coefficients • hypothesis tests

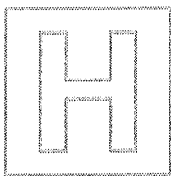
ASSUMED KNOWLEDGE

Content from O-Level Additional Mathematics	
ALGEBRA	
A1	Quadratic functions; Equations and inequalities - Finding the maximum or minimum value of a quadratic function using the method of completing the square - Conditions for a quadratic equation to have: - two real roots - two equal roots - no real roots - Conditions for $ax^2 + bx + c$ to be always positive (or always negative) - Solving simultaneous equations in two variables by substitution, with one of the equations being a linear equation
A2	Surds - Four operations on surds, including rationalising the denominator - Solving equations involving surds.
A3	Polynomials and partial fractions - Multiplication and division of polynomials - Use of remainder and factor theorems - Partial fractions with cases where the denominator is not more complicated than: $(ax + b)(cx + d)$ $(ax + b)(cx + d)^2$ $(ax + b)(x^2 + c^2)$
A4	Exponential and logarithmic functions - Exponential and logarithmic functions a^x, e^x, $\log_a x$, $\ln x$ and their graphs, including: - laws of logarithms - equivalence of $y = a^x$ and $x = \log_a y$ - change of base of logarithms - Simplifying expressions and solving simple equations involving exponential and logarithmic functions
GEOMETRY AND TRIGONOMETRY	
B5	Coordinate geometry in two dimensions Coordinate geometry of the circle with the equation in the form $(x - a)^2 + (y - b)^2 = r^2$ or $x^2 + y^2 + 2gx + 2fy + c = 0$
B6	Trigonometric functions, identities and equations - Six trigonometric functions, and principal values of the inverses of sine, cosine and tangent - Trigonometric equations and identities (see List of Formulae) - Expression of $a \cos \theta + b \sin \theta$ in the forms $R \sin(\theta \pm \alpha)$ and $R \cos(\theta \pm \alpha)$

Content from O-Level Additional Mathematics

CALCULUS

C7	Differentiation and integration • Derivative of $f(x)$ as the gradient of the tangent to the graph of $y = f(x)$ at a point • Derivative as rate of change • Derivatives of x^n for any rational n, $\sin x$, $\cos x$, $\tan x$, e^x and $\ln x$, together with constant multiples, sums and differences • Use of Chain Rule • Derivatives of products and quotients of functions • Increasing and decreasing functions • Stationary points (maximum and minimum turning points and points of inflexion) • Use of second derivative test to discriminate between maxima and minima • Connected rates of change • Maxima and minima problems • Integration as the reverse of differentiation • Integration of x^n for any rational n, e^x, $\sin x$, $\cos x$, $\sec^2 x$ and their constant multiples, sums and differences • Integration of $(ax + b)^n$ for any rational n, $\sin(ax + b)$, $\cos(ax + b)$ and $e^{ax + b}$
----	---



MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level

List MF27

LIST OF FORMULAE

AND

RESULTS

for Mathematics and Further Mathematics

For use from 2025 in all papers for the H1, H2 and H3 Mathematics and
H2 Further Mathematics syllabuses.

This document has 8 pages.



Singapore Examinations and Assessment Board



CAMBRIDGE
International Education

Algebraic series

Binomial expansion:

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \binom{n}{3} a^{n-3}b^3 + \dots + b^n, \text{ where } n \text{ is a positive integer}$$

$$\text{and } \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Maclaurin expansion:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!} x^r + \dots \quad (|x| < 1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots \quad (\text{all } x)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \dots \quad (\text{all } x)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^r x^{2r}}{(2r)!} + \dots \quad (\text{all } x)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{r+1} x^r}{r} + \dots \quad (-1 < x \leq 1)$$

Partial fractions decomposition

Non-repeated linear factors:

$$\frac{px+q}{(ax+b)(cx+d)} = \frac{A}{(ax+b)} + \frac{B}{(cx+d)}$$

Repeated linear factors:

$$\frac{px^2+qx+r}{(ax+b)(cx+d)^2} = \frac{A}{(ax+b)} + \frac{B}{(cx+d)} + \frac{C}{(cx+d)^2}$$

Non-repeated quadratic factor:

$$\frac{px^2+qx+r}{(ax+b)(x^2+c^2)} = \frac{A}{(ax+b)} + \frac{Bx+C}{(x^2+c^2)}$$

Trigonometry

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2 \cos^2 A - 1 \equiv 1 - 2 \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

Principal values:

$$-\frac{1}{2}\pi \leq \sin^{-1} x \leq \frac{1}{2}\pi \quad (|x| \leq 1)$$

$$0 \leq \cos^{-1} x \leq \pi \quad (|x| \leq 1)$$

$$-\frac{1}{2}\pi < \tan^{-1} x < \frac{1}{2}\pi$$

Derivatives

$f(x)$	$f'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\sec x$	$\sec x \tan x$

Integrals

(Arbitrary constants are omitted; a denotes a positive constant.)

$f(x)$	$\int f(x) \, dx$	
$\frac{1}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$	
$\frac{1}{\sqrt{a^2 - x^2}}$	$\sin^{-1}\left(\frac{x}{a}\right)$	$(x < a)$
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$	$(x > a)$
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right)$	$(x < a)$
$\tan x$	$\ln(\sec x)$	$(x < \frac{1}{2}\pi)$
$\cot x$	$\ln(\sin x)$	$(0 < x < \pi)$
$\operatorname{cosec} x$	$-\ln(\operatorname{cosec} x + \cot x)$	$(0 < x < \pi)$
$\sec x$	$\ln(\sec x + \tan x)$	$(x < \frac{1}{2}\pi)$

Vectors

The point dividing AB in the ratio $\lambda : \mu$ has position vector $\frac{\mu \mathbf{a} + \lambda \mathbf{b}}{\lambda + \mu}$

Vector product:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

Applications of definite integrals

Arc length of a curve defined in cartesian form: $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$

Surface area of revolution about the x -axis for a curve defined in cartesian form:

$$\int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

Functions of two variables

Quadratic approximation of f at (a, b) :

$$\begin{aligned} f(x, y) \approx & f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ & + \frac{1}{2} f_{xx}(a, b)(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{1}{2} f_{yy}(a, b)(y - b)^2 \end{aligned}$$

Numerical methods

Trapezium rule (for single strip): $\int_a^b f(x)dx \approx \frac{1}{2}(b-a)[f(a) + f(b)]$

Simpson's rule (for two strips): $\int_a^b f(x)dx \approx \frac{1}{6}(b-a) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$

The Newton-Raphson iteration for approximating a root of $f(x) = 0$:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)},$$

where x_1 is a first approximation.

Euler Method with step size h :

$$y_2 = y_1 + hf(x_1, y_1)$$

Improved Euler Method with step size h :

$$u_2 = y_1 + hf(x_1, y_1)$$
$$y_2 = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, u_2)]$$

Standard discrete distributions

Distribution of X	$P(X = x)$	Mean	Variance
Binomial $B(n, p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson $Po(\lambda)$	$e^{-\lambda} \frac{\lambda^x}{x!}$	λ	λ
Geometric $Geo(p)$	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

Standard continuous distribution

Distribution of X	p.d.f.	Mean	Variance
Exponential	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Sampling and testing

Unbiased estimate of population variance:

$$s^2 = \frac{n}{n-1} \left(\frac{\Sigma(x - \bar{x})^2}{n} \right) = \frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Regression and correlation

Estimated product moment correlation coefficient:

$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{\left\{ \Sigma(x - \bar{x})^2 \right\} \left\{ \Sigma(y - \bar{y})^2 \right\}}} = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right) \left(\Sigma y^2 - \frac{(\Sigma y)^2}{n} \right)}}$$

Estimated regression line of y on x :

$$y - \bar{y} = b(x - \bar{x}), \quad \text{where } b = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(x - \bar{x})^2}$$

WILCOXON SIGNED RANK TEST

P is the sum of the ranks corresponding to the positive differences,

Q is the sum of the ranks corresponding to the negative differences,

T is the smaller of P and Q .

For each value of n the table gives the **largest** value of T which will lead to rejection of the null hypothesis at the level of significance indicated.

Critical values of T

	Level of significance			
	0.05	0.025	0.01	0.005
One				
Two	0.1	0.05	0.02	0.01
$n = 6$	2	0		
7	3	2	0	
8	5	3	1	0
9	8	5	3	1
10	10	8	5	3
11	13	10	7	5
12	17	13	9	7
13	21	17	12	9
14	25	21	15	12
15	30	25	19	15
16	35	29	23	19
17	41	34	27	23
18	47	40	32	27
19	53	46	37	32
20	60	52	43	37

Mathematical Results

AM-GM inequality:

For any nonnegative real numbers $x_1, x_2, \dots, x_n$,

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n},$$

where the equality holds if and only if $x_1 = x_2 = \dots = x_n$.

Cauchy-Schwarz inequality:

For any real numbers $u_1, u_2, \dots, u_n$ and $v_1, v_2, \dots, v_n$,

$$\left(\sum_{i=1}^n u_i v_i \right)^2 \leq \left(\sum_{i=1}^n u_i^2 \right) \left(\sum_{i=1}^n v_i^2 \right),$$

where the equality holds if and only if there exists a nonzero constant k such that $u_i = k v_i$ for all $i = 1, 2, \dots, n$.

Triangle inequality:

For any real numbers $x_1, x_2, \dots, x_n$,

$$|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n|,$$

where the equality holds if $x_1, x_2, \dots, x_n$ are all nonnegative.

Inclusion-Exclusion Principle:

For any subsets $A_1, A_2, \dots, A_n$ of a set,

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= |A_1| + |A_2| + \dots + |A_n| \\ &\quad - [|A_1 \cap A_2| + |A_1 \cap A_3| + \dots + |A_{n-1} \cap A_n|] \\ &\quad + [|A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + \dots + |A_{n-2} \cap A_{n-1} \cap A_n|] \\ &\quad \vdots \\ &\quad + (-1)^{n-1} |A_1 \cap A_2 \dots \cap A_{n-1} \cap A_n| \end{aligned}$$

This booklet is the property of

SINGAPORE EXAMINATIONS AND ASSESSMENT BOARD



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Summary and Tutorial)

Topic: Equations and Inequalities

Summary for System of Linear Equations

This topic involves

1. Understand the given question
2. Formulate the equations
3. Use of GC to solve the equations

Recap on the use of GC:

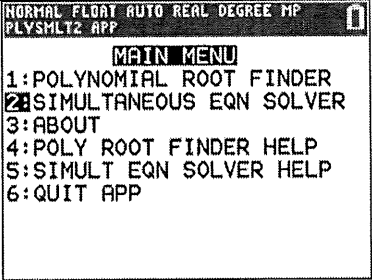
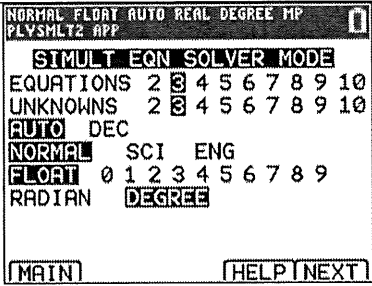
Consider the following equations:

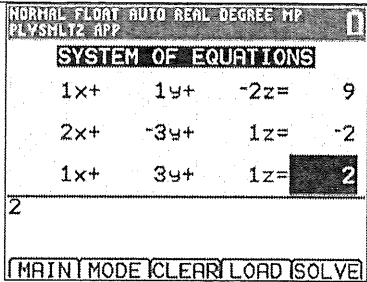
$$x + y - 2z = 9$$

$$2x - 3y + z = -2$$

$$x + 3y + z = 2$$

$$x = 2, y = 1, z = -3.$$

1. Press [apps] and select PlySmlt2. 2. Select 2: Simultaneous Eqn Solver.	
3. Adjust the settings as depicted on the screen to solve 3 equations with 3 unknowns. 4. Select NEXT by pressing the s button.	

5. Enter the coefficients into the equations. 6. Press SOLVE the system of equations.	
7. Read off the answers. Hence $x = 2, y = 1, z = -3$.	